

DEEPAK GHIMIRE, Ph.D.

Computer Vision & AI/ML · Edge AI · 2D/3D Perception & Robotics · Model Optimization

@ghmdeepak@gmail.com 010-6599-1928 Jeonju, South Korea
ghimiredeepak.com.np ghimire-deepak ghimiredhikura 0000-0001-8940-8739



Experience: 15+ Years (10+ after PhD)

Senior Research Engineer

Korea Electronics Technology Institute (KETI)

- Jul 2024 – Present Jeonju, Korea
- Lead AI and vision R&D for edge and autonomous systems in robotics, agriculture, and surveillance.
 - Develop delta-robot vision for pick-and-place and real-time quality inspection.
 - Build autonomous tractor guidance using GPS, IMU, LiDAR, and cameras for paddy fields.

Research Professor

Soongsil University

- Jan 2022 – Jun 2024 Seoul, Korea
- Developed efficient deep learning solutions for edge computing, focusing on one-shot structured pruning to optimize model performance and resource utilization.
 - Taught Computer Vision & Deep Learning; supervised graduate students in AI/ML research.

AI / Computer Vision Engineer (Freelance)

BeEmotion.ai (formerly NVISO.ai)

- Feb 2018 – Apr 2022 Remote via Upwork Inc.
- Benchmarked DNN backbones, face detection, and facial emotion models using state-of-the-art inference tools on Jetson & Raspberry Pi.
 - Contributed to an in-car Interior Monitoring System (IMS) for driver state and occupant behavior.

Postdoctoral Researcher

Korea Electronics Technology Institute (KETI)

- Jul 2014 – Feb 2020 Seoul, Korea
- Developed vision-based ADAS algorithms (driver state monitoring, forward collision warning, lane departure warning) for dashcams & mobile.
 - Developed algorithms for pedestrian/vehicle detection, bird's-eye-view 3D perception and stereo depth estimation.

Earlier Roles

Jeonbuk National Univ., Simon Fraser Univ., Lumbini Eng. College

- 2007 – 2014 Korea, Canada, Nepal
- Graduate Research Assistant at Jeonbuk National University, Korea (Mar 2009 – Aug 2014): face detection, facial expression analysis, driver monitoring, traffic surveillance, and object tracking.
 - Visiting PhD Scholar at Simon Fraser University (SFU), Canada (Feb 2014 – Apr 2014): geometric feature extraction and facial expression analysis at the Vision and Media Lab.
 - Teaching Faculty at Lumbini Engineering College, Nepal (Aug 2007 – Nov 2008): C++, Computer Graphics, Digital Signal Processing, and Image Processing with hands-on lab instruction.

Education

Ph.D. in Computer Science & Engineering

[CGPA: 97.2/100]

Jeonbuk National University

- Mar 2011 – Aug 2014 Jeonju, Korea
- Thesis:** Recognition of Facial Expressions Based on Salient Texture and Geometric Features.

M.Sc. in Computer Science & Engineering

[CGPA: 96.5/100]

Jeonbuk National University

- Mar 2009 – Feb 2011 Jeonju, Korea
- Thesis:** A Robust Face Detection Method Based on Skin Color and Edges.

Technical Skills

- AI / Vision:** 2D & 3D perception, object detection, ADAS, tracking, model compression, structured pruning.
- Frameworks:** PyTorch, TensorFlow, TensorRT, OpenVINO, OpenCV, Open3D, MMDetection3D, NCNN, and MNN.
- Edge:** Jetson Nano/Xavier/Orin, Raspberry Pi, and ARM systems.
- Programming:** Python, C/C++; experience with MATLAB and Java.
- Robotics / Tools:** ROS1/ROS2, Qt, Visual Studio, Linux, Git, real-time inference optimization.
- Research & Deployment:** Peer-reviewed publications, edge-AI benchmarking, real-time system integration.

Others

Citation Metrics as of August 13, 2026

Platform	Articles	h-index	Citations
Google Scholar	53	20	1981
ResearchGate	54	19	1615
Semantic Scholar	51	17	1375
Scopus	40	18	1343

Languages

Nepali (Native)	■■■■■■■■■■
English (C1 - Advanced)	■■■■■■■■■■
Korean (Fluent)	■■■■■■■■■■
Hindi (Fluent)	■■■■■■■■■■

Proficiency levels are self-assessed.